Ligjerata 9d

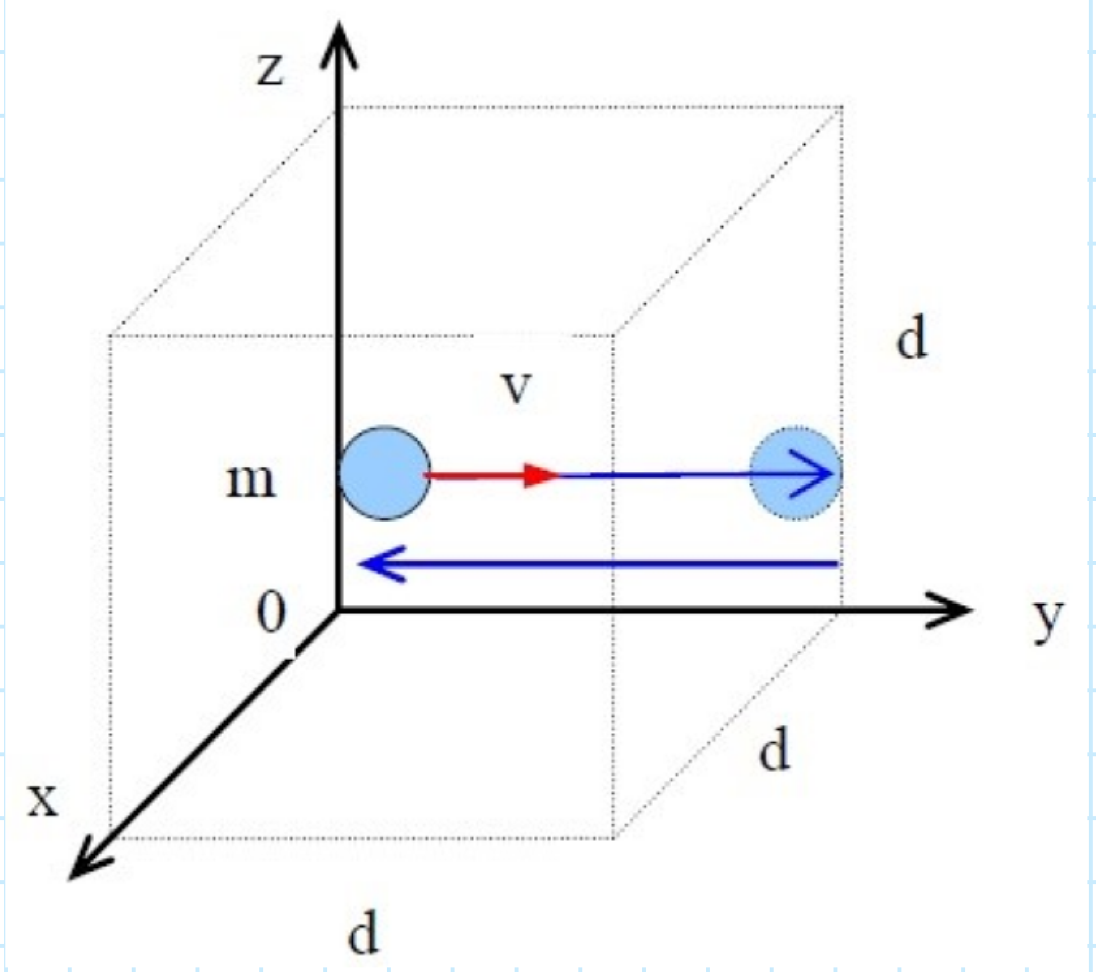
Teoria kinetike molekulare e gazit

Shtypja e molekulave në muret e enës

$$\Delta v = v_1 - v_2 = v_y - (-v_y) = 2v_y$$

$$mv - m(-v) = \underline{\underline{2mv}}$$

$$\Delta F \cdot \Delta t = 2mv$$



$$\Delta F = \frac{2mv}{\Delta t}$$

$$\Delta t = \frac{2d}{v}$$

$$\Delta F = \frac{\cancel{2}mv}{\frac{\cancel{2}d}{v}} = \frac{mv^2}{d}$$

$$F = \frac{mv_1^2}{d} + \frac{mv_2^2}{d} + \dots + \frac{mv_n^2}{d}$$

$$F_m = \frac{n'm}{d} \underbrace{\frac{v_1^2 + v_2^2 + \dots + v_{n'}^2}{n'}}_{v_m^2}$$

$$v_m^2 = \frac{v_1^2 + v_2^2 + \dots + v_{n'}^2}{n'}$$

$$F_m = \frac{n' m v_m^2}{d}$$

$$n' = \frac{1}{3} n$$

$$d^2 = S$$

$$F_m = \frac{1}{3} \frac{n}{d} m v_m^2 \Big/ : d^2$$

$$d^3 = V$$

$$\frac{F_m}{d^2} = \frac{1}{3} \frac{n}{d^3} \cdot m v_m^2$$

$$\frac{n}{V} = n_0$$

$$\boxed{p = \frac{F_m}{d^2}}$$

$$v_m^2 = \langle v^2 \rangle$$

$$\boxed{p = \frac{1}{3} n_0 m v_m^2}$$

$$\boxed{p = \frac{1}{3} n_0 m \langle v^2 \rangle}$$ shtypja e gazit ideal në muret e brendshme të enës.

Kur kemi gaz të përzier.

$$n_0 = n_{01} + n_{02} + \dots$$

$$p = \frac{1}{3} n_0 m v_m^2 \Big/ \cdot \frac{2}{2}$$

$$p = \frac{1}{3} n_{01} m v_m^2 + \frac{1}{3} n_{02} m v_m^2 \dots$$

$$p = \frac{2}{3} n_0 E_k$$

$$\boxed{E_k = \frac{m v_m^2}{2}}$$

$$p_1 = \frac{2}{3} n_{01} E_k$$

$$p_2 = \frac{2}{3} n_{02} E_k$$

$$\vdots$$

$$p = p_1 + p_2 + \dots$$ Ligji i Daltonit